

Aishwarya Govindappa Mahadev

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PROFESSIONAL EXPERIENCE

Senior Software Engineer (Product & Platform Engineering),

09/2023 – Present | Bangalore, India

Altimetrik (Client: Airbnb)

- Extensively worked in the AI Search and Knowledge Graph domain. Became the **POC for caching in AI search services** by building a shared AI search library that wraps LLM callers with a reusable caching decorator, standardizing LLM call patterns and cached responses with request-level flags to safely bypass cache when needed.
- Introduced a unified Redis client abstraction in the search platform, replacing legacy Redisson/Jedis usage with a shared KV store backed by the unified client, enabling single round-trip bulk reads, deleting duplicate client code, simplifying dependency wiring, and reducing Redis latency and maintenance overhead across multiple production caches.
- Implemented and tuned Redis-backed, multi-layer caching by provisioning AWS ElastiCache Redis clusters via Terraform-based IaC workflows across environments for Search and Knowledge Infrastructure services, integrating cache usage at endpoint, guardrail, and LLM-call levels for two high-cost search endpoints reducing Redis access latency and improved endpoint P99 latency, while adding feature-flag-driven controls for cache policies and TTLs so teams could gradually roll out caching, tune behaviour in production, and instantly disable problematic configurations without redeploying code.
- Added new **regex-based guardrail/jailbreak patterns** to detect discriminatory AD&E phrases combining protected characteristics with hosting-related terms (for example, “run by black host”), preventing such harmful queries from bypassing safety checks and reaching system internals. Migrated services to newer open-source and hosted model versions as needed, including GPT-4 to GPT-5 transitions where applicable.
- Increased **observability and testability** of LLM-powered endpoints by adding end-to-end debug logging and config-driven switches for query parsing and LLM responses, cutting time-to-diagnose production issues and enabling safer, incremental feature rollouts for AI features.
- Delivered first-of-its-kind **integration coverage** by building an end-to-end testing framework for Knowledge Graph pipelines (Flink, Gremlin, JanusGraph, Mussel), enabling scenario-driven validation of complex data flows before production releases.
- Strengthened **system reliability** of Knowledge Graph services by implementing comprehensive unit, integration, regression, cron, and E2E test suites, reducing production defects and improving runtime stability across core graph workloads.
- Proactively **migrated Azure OpenAI workloads** for query-understanding services from deprecated GPT-4x models to gpt-5 variants and Facade v2 API endpoints ahead of Azure’s retirement deadlines, performing evaluation, testing, and config updates to ensure zero service downtime and full compliance with platform deprecations.
- Modernised **high-volume Spark data pipelines** by migrating and optimising legacy jobs and updating Airflow DAGs, improving job runtime performance and stability while adding test coverage to support smooth Spark version upgrades.
- Reduced deployment risk by designing optimised CI/CD pipelines in Spinnaker that ran tests across environments, with mocking fixtures for service dependencies, proactive service alerts, and a standardised shadow testing workflow validating new changes against live production behaviour before rollout, improving confidence in releases for high-traffic services.
- Improved **build efficiency and maintainability** of monorepo Java/Scala services by refactoring legacy code paths and simplifying module boundaries, resulting in faster CI builds, lower maintenance overhead, and more predictable deployments.

Advanced Software Engineer, Zuhlke (Client: Credit Suisse)

08/2021 – 08/2022 | Singapore, Singapore

- Built and delivered **production-grade financial features** for Credit Suisse's wealth management application, working with product owners and business analysts to translate regulatory and business requirements into scalable, secure backend services used by wealth management clients.
- Co-developed and **owned three RESTful microservices** powering critical portfolio, trading, and advisory workflows, designing APIs, implementing business logic, and ensuring secure, low-latency integration with upstream and downstream systems across the platform.
- Improved **system resilience** of a significant trading workflow by designing and implementing a fallback mechanism that removed single points of failure and reduced dependency on third-party APIs, keeping core trading functionality available during external outages.
- Enhanced **software quality** by implementing comprehensive unit and integration test suites for core services, increasing automated test coverage and reducing production defects and release rollback frequency.
- Ensured stable **production operations** by performing root-cause analysis, debugging incidents, and running production sanity checks, enabling smoother feature rollouts and minimizing client-facing disruptions in wealth management workflows.
- Delivered improvements to **customer-facing UI** components by gathering user and stakeholder requirements, aligning with business expectations, and implementing intuitive, responsive interfaces that better supported advisors' daily workflows.
- Led **code reviews**, authored test plans, and supported production deployments while mentoring junior engineers and maintaining clear technical documentation, raising overall code quality and team execution standards.

Software Engineer, Garmin

10/2019 – 12/2020 | Olathe, USA

- Contributed to the development of BMW's Manufacturing Execution System (MES) for infotainment unit production, working within a high-availability environment where real-time data accuracy and system reliability were critical.
- Designed and implemented **RESTful APIs** following best practices in architecture, security, versioning, and performance, backed by unit/integration tests and automated CI/CD pipelines, enabling safe and repeatable deployments of MES functionality. Designed, set up, and **maintained** Oracle database schemas and tables for these microservices.
- Owned and maintained multiple **production-critical MES services** from inception through deployment, supporting feature development, infrastructure migrations, and long-term stability for manufacturing operations.
- Collaborated with manufacturing engineers and process experts to integrate MES services into end-to-end production flows, and provided **knowledge transfer** to team members, accelerating onboarding and reducing integration issues.
- Led daily troubleshooting of **real-time manufacturing discrepancies**, conducted code reviews, refined API designs, and built Grafana dashboards for service-level monitoring, significantly improving visibility into production metrics and system health.
- Facilitated **Scrum ceremonies** and produced detailed technical documentation, becoming the go-to technical resource for new team members and helping enforce consistent engineering and delivery practices across the team.

EDUCATION

MSc in Computer Science, Texas A&M University

2018 – 2019 | Commerce, USA

SKILLS

Languages — Java, Scala, Python

Backend & Frameworks — Spring Boot, Microservices, Monoliths, API Design (RESTful, GraphQL), MVC, SOA

Cloud & Infrastructure — AWS, OpenShift, Pivotal Cloud Foundry, Docker, Kubernetes, Vault, Terraform

Data & Storage — Oracle DB, MySQL, Key-value stores (e.g. Redis on AWS ElastiCache), JanusGraph

Data Platforms & Streaming: — Spark, Flink, Airflow, Kafka, Superset

Build & CI/CD: — Maven, Gradle, Bazel, Buildkite, EngFlow, Jenkins, GitLab CI, TeamCity, Spinnaker, Git

Practices & Methodologies — TDD, CI/CD, Agile, Scrum, shadow testing and config-driven regression testing for backend services

AI & LLMs: — Prompt engineering, Generative AI, working with LLMs (GitHub Copilot, GPT, Claude)

Certificates

Generative AI for Everyone - DeepLearning.AI 
(Sept 2025)

Prompt Engineering (Airbnb, DeepLearning.AI, Altimetrik) 
(Sept 2025)

Introduction to Responsible AI - Google Cloud 
(Sept 2025)

Trustworthy Generative AI - Vanderbilt University 
(Sept 2025)

IN THE BACKGROUND

Professional Athlete

- Represented India at the Commonwealth Youth Games and various Asian Athletic Championships
- Awarded the best sportsperson by Karnataka Olympic Association in 2008. Won 34 National Level medals at various athletic events.

Academic Tutor

- Volunteered as a teacher and supervisor for a hostel run by the Government of India with 108 female students of various age groups. Coached the inmates in academic subjects and personal development activities.
- Entered the final round of the Rhodes Scholarships selections in 2015.